From Functional Data to Smooth Functions

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Overview

- We need a flexible method for constructing functions from noisy discrete data.
- The method should be able to reproduce any feature that interests us in a function, no matter how complicated.
- The computation should be reasonably fast, even when tens or hundreds of thousands of discrete values are available.
- In this class, we consider the most popular technique, *basis function expansions*.

Overview

- We describe two basis function systems in detail:
  - Fourier bases
  - B spline bases
- as well as some other important systems.
- We also ask about how to estimate derivatives,
- including a bad idea.

Basis representation

- A basis function system is a set of K known functions $\phi_k(t)$ that are:
  - linearly independent of each other,
  - can be extended to include any number K in the system.
- A function $x(t)$ is constructed as a linear combination of these basis functions:

$$x(t) = \sum_{k=1}^{K} c_k \phi_k(t).$$

- If vector $\mathbf{c}$ contains the coefficients, and the vector $\boldsymbol{\phi}$ contains the basis functions, then

$$x(t) = \mathbf{c}' \boldsymbol{\phi}(t).$$

Derivatives in basis function systems

- In principle, computing derivatives is easy:

$$D^m x(t) = \sum_{k=1}^{K} c_k D^m \phi_k(t).$$

- but not all basis functions have derivatives that behave reasonably, or can even be calculated.

The monomial basis

- Polynomials are perhaps the oldest and best known basis function expansion.
- A polynomial is the form:

$$x(t) = \sum_{k=1}^{K} c_k t^{k-1}.$$

- The basis functions are the monomials: $1, t, t^2, t^3, \ldots$
- Polynomials can work fine for simple problems only requiring $K = 5$ or so, but have severe problems tracking sharp localized features, and can run into computational problems for unequally spaced data.

Derivatives of polynomials

- Derivative estimation is a big problem for polynomials because their derivatives become less and less complex, the higher the order of derivative.
- For a polynomial of degree m, the derivative of order $m + 1$ is zero.
- But in most realworld systems, derivatives become more complex as the order of the derivative increases.

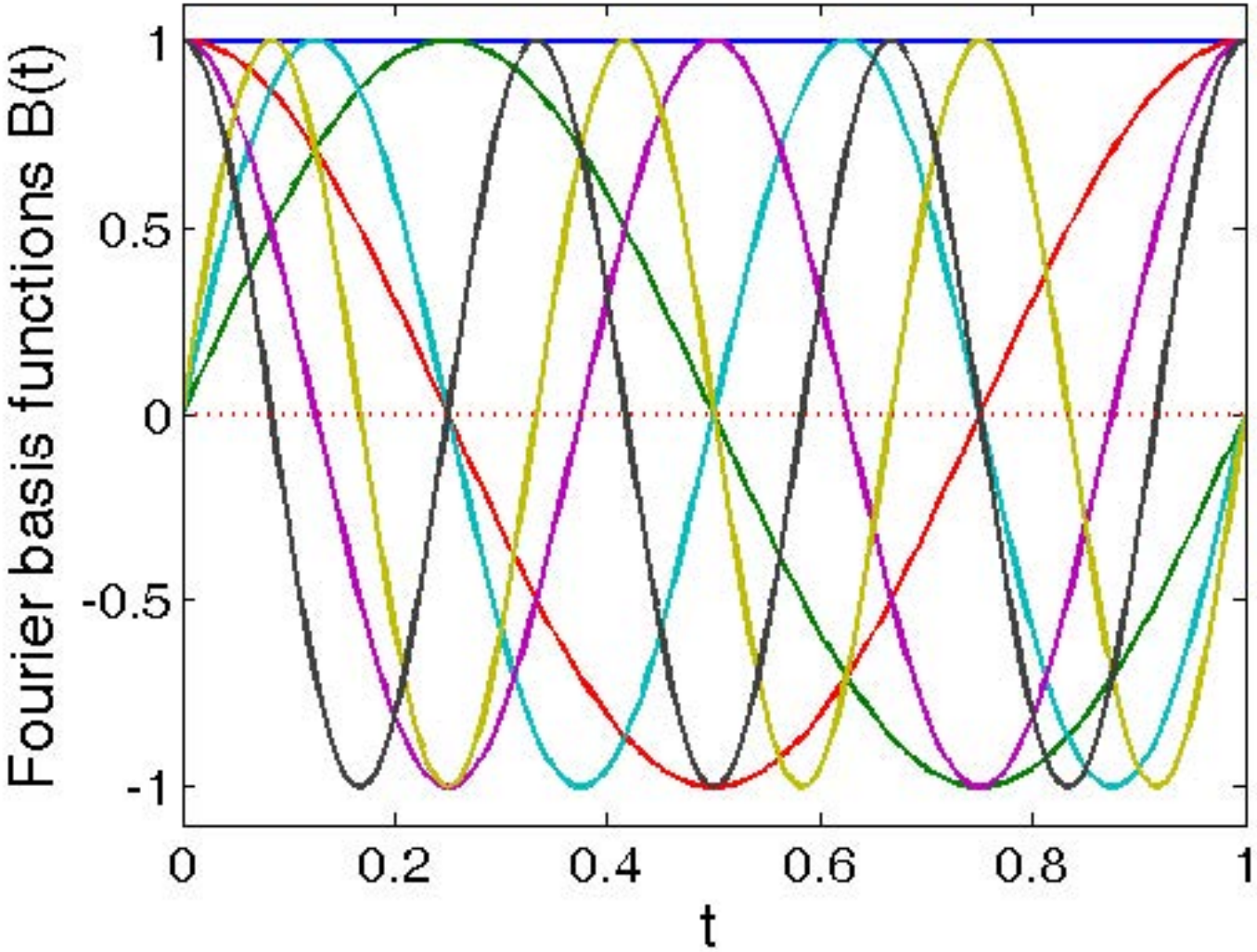
Fourier basis functions

- The basis functions are sine and cosine functions of increasing frequency:

$$1, \sin(\omega t), \cos(\omega t), \ldots, \sin(m\omega t), \cos(m\omega t).$$

- The constant ω defines the period of oscillation of the first sine/cosine pair. That is, $\omega = 2\pi/P$ where P is the period.
- $K = 2M + 1$ where M is the largest number of oscillations in period P that are required.

Fourier basis functions



Advantages of Fourier basis functions

- Fourier bases were the only alternative to monomial bases until the middle of the 20th century.
- They have excellent computational properties, especially if the times of observation are equally spaced.
- They are natural for describing data which are periodic, such as the annual weather data, gait cycle data and so on.
- Their periodicity is a problem, however, for nonperiodic data, such as the growth curves.
- But the Fourier basis is still the first choice in many fields, such as signal analysis, even when the data are not periodic.

Derivatives of Fourier bases

- Computing derivatives is easy since

$$D\sin(\omega t) = \omega\cos(\omega t), D\cos(\omega t) = -\omega\sin(\omega t).$$

- We say that this system is closed under differentiation: the derivative of a Fourier series expansion is also a Fourier series expansion.
- The Fourier series is infinitely differentiable.

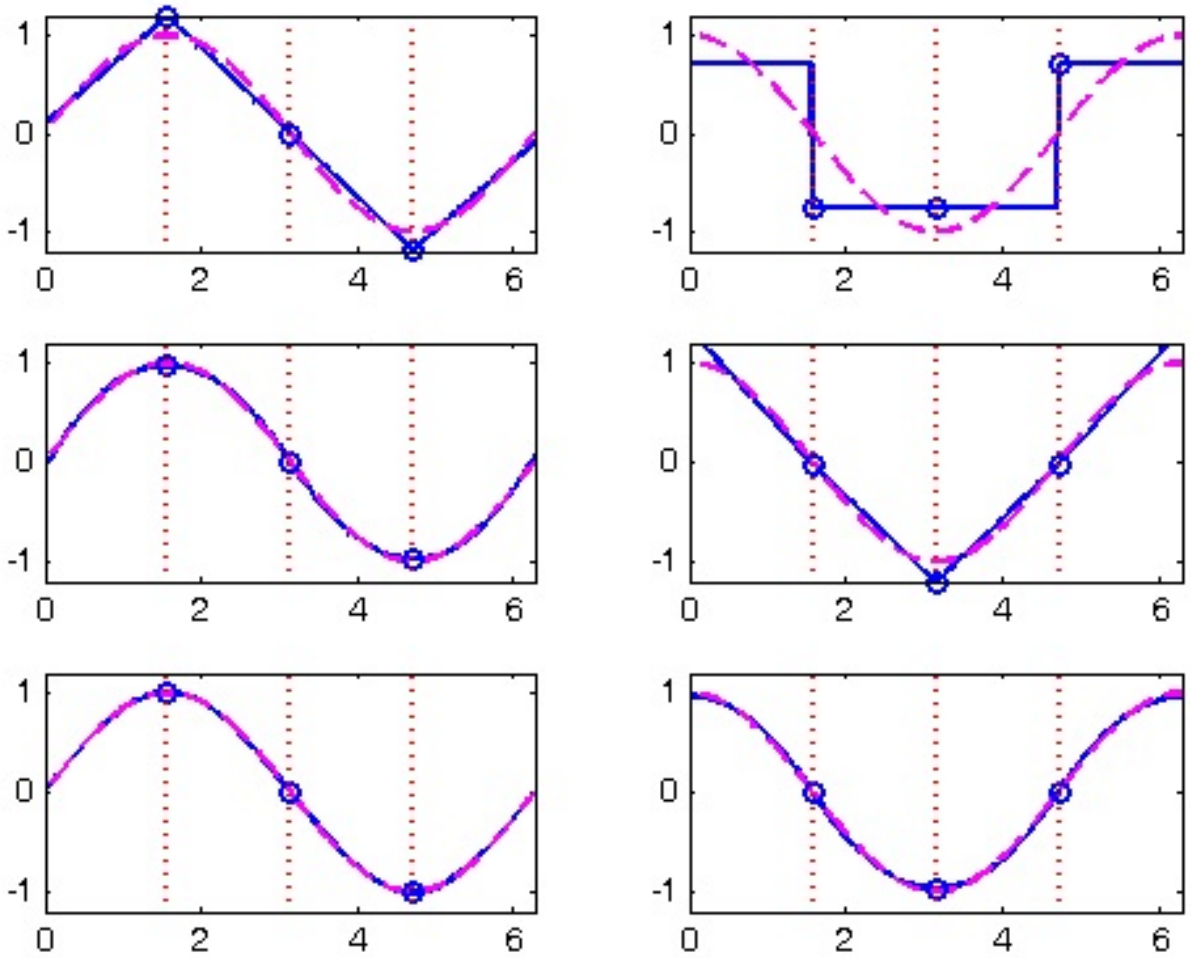
The splines

- Splines are polynomial segments joined end-to-end.
- The segments are constrained to be smooth at the join. The values of t at which adjacent segments are joined are called *knots*.
- The order $m(\text{order} = \text{degree} + 1)$ of the polynomial segments and the location of the knots define the spline basis system.

An example of spline functions

- The following figure shows splines of three orders, each with three knot values.
- The splines are defined so as to offer the best fit to a sine function, shown in the left panels.
- How well the derivatives of these splines fit the derivative of the sine, the cosine, is shown in the right panels.

An example of spline functions



Derivatives with splines

- Because splines are constructed from polynomials, computing their derivative at any point between two knots is simple. There, the highest nontrivial order of derivative is $m-1$ for order m splines.
- At a knot, it is usual to require that the derivatives up to order $m-2$ also join. That is, the derivative of order $m-2$ of a spline function is usually continuous.
- The most popular choice of order is 4, implying continuous second derivatives. The second derivatives have straight line segments.

Spline functions and DFs

- How can we quantify the flexibility of a spline function of order m?
- In the usual case, there are $m-1$ constraints on the adjacent polynomials, corresponding to the requirement that $m-2$ derivatives plus the function values are required to match at the knot.
- Given the first segment, with m degrees of freedom, this means that we gain one degree of freedom with each knot to the right of the first segment.
- The total number of degrees of freedom is

$$\text{order } m + \text{number of interior knots.}$$

How are knots chosen?

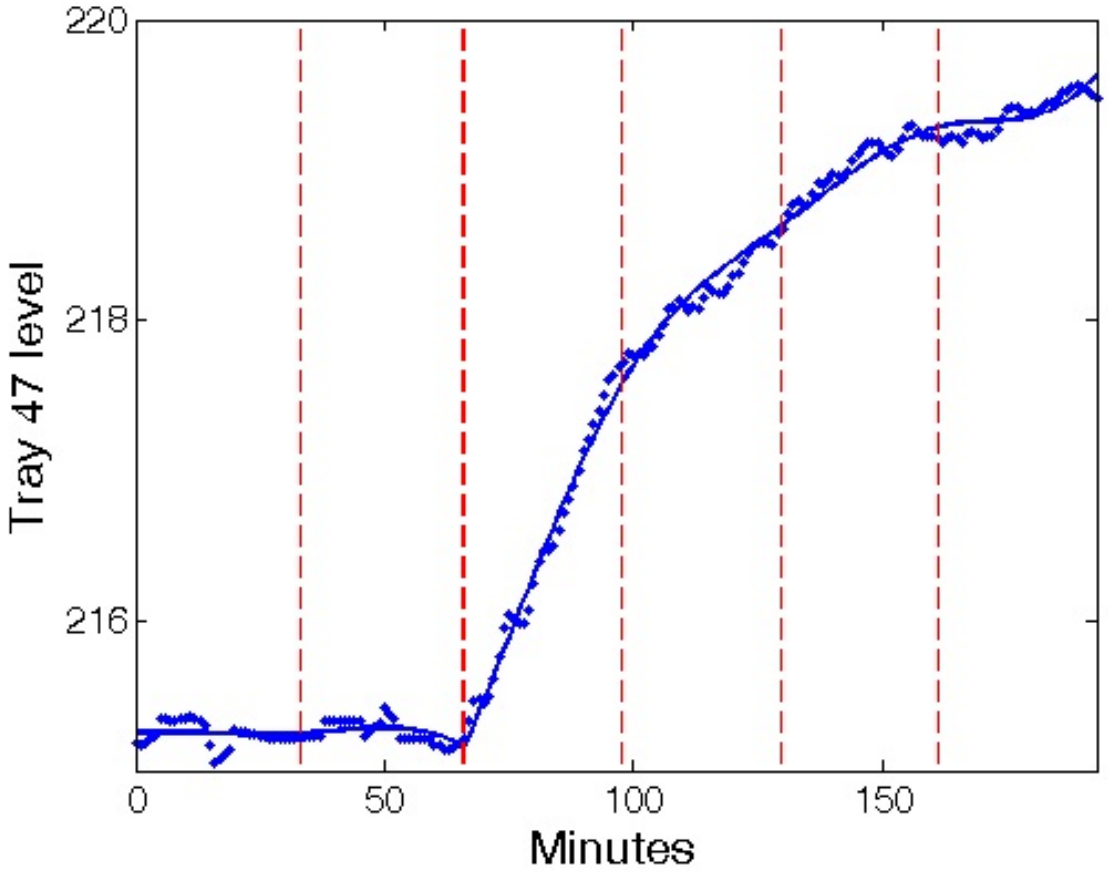
- Knots are often spaced equally.
- But two important rules should be observed in placing knots:
  - Place more knots where you know there is strong curvature, and fewer where the function changes slowly.
  - But be sure that there is at least one data point in any interval.
- Later, we will consider placing a knot at each point of observation.

Coincident knots in splines

- Sometimes we need less smoothness at a specific point.
- For example, we will see problems where a function needs to be continuous at a point, but its derivative is discontinuous.
- When multiple knots are placed at the same point, the convention is that a spline loses one derivative for each additional knot.
- An order 4 spline with 3 coincident knots is continuous at that point, but does not have a first derivative.
- An order 4 spline with 4 coincident knots is discontinuous at that point.

An example: refinery data

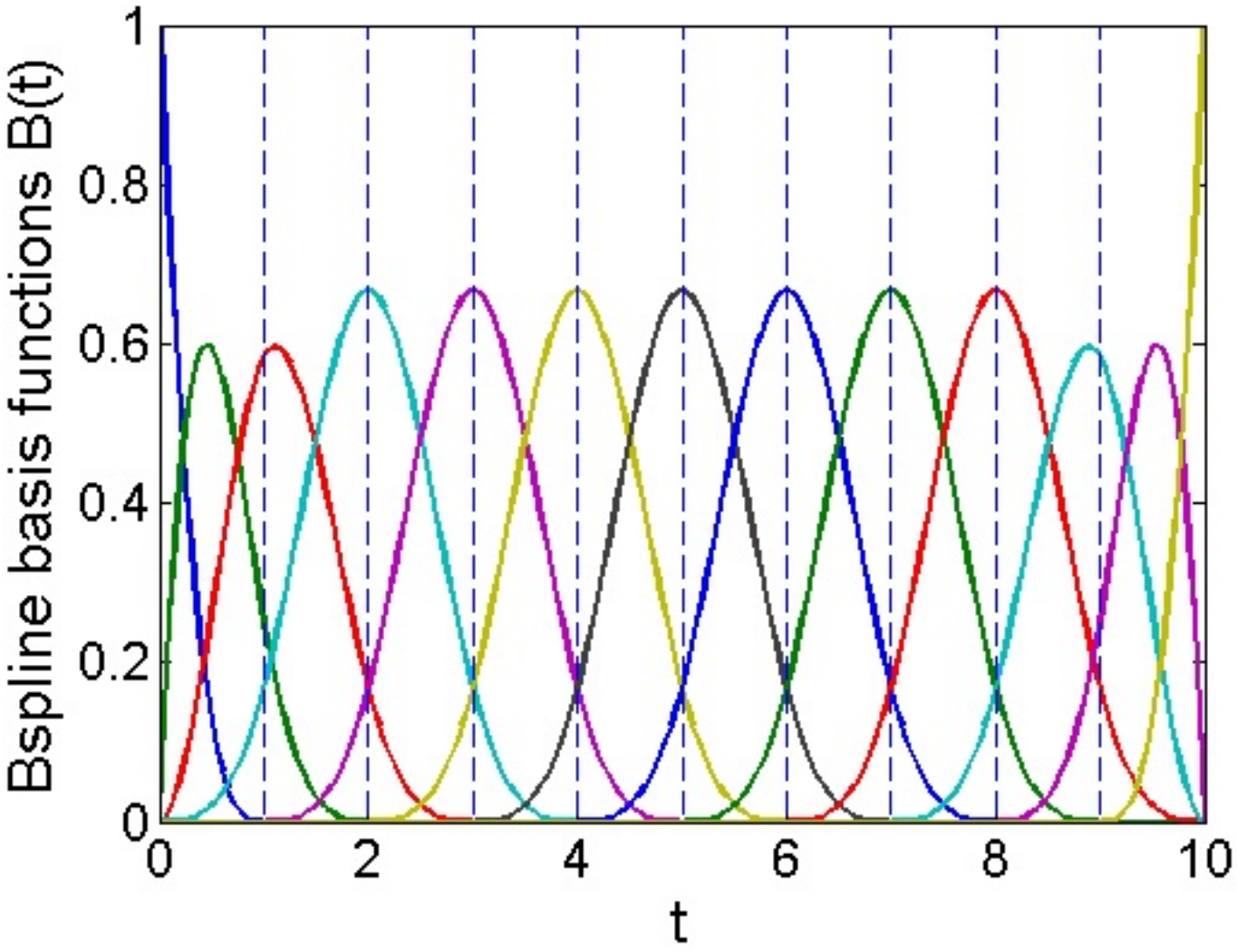
There are three coincident knots at the second location for the refinery data to permit a discontinuous first derivative.



The B-spline basis system

- Any spline function with K degrees of freedom can be expressed as a linear combination of K basis spline functions.
- Among many possibilities, the B-spline system, developed in the 1940s, is the most popular.
- B-spline basis functions are themselves spline functions.
- Any B-spline basis function is positive over at most m adjacent intervals.
- This ensures that computation is fast for even tens of thousands of basis functions.

13 order 4 B-spline basis functions



Power basis and exponential basis systems

- **Power basis**: $t^{\lambda_1}, t^{\lambda_2}, \ldots$, where the powers are distinct but not necessarily integers or even positive.
- **Exponential basis**: $e^{\lambda_1 t}, e^{\lambda_2 t}, \ldots$, where the λ's are distinct.

Wavelet basis functions

- A recent development, wavelet bases combine some of the advantages of both Fourier and B-spline bases.
- They are especially good at tracking sharp highly localized features,
- and separating a signal into components which reflect both specific frequencies and specific locations on the t-axis. Because of their computational efficiency, they are often used for image compression.
- For example, the FBI uses wavelets to store fingerprint information.

Empirical basis functions

- We will look at *functional principal components analysis* later.
- This is essentially a method for estimating orthogonal basis functions from functional data that capture as much of the variation as possible given a fixed number of basis functions K.

Designer or customized basis functions

- Later, when we come to differential equation models, we will see how to tailor a basis system to the known characteristics of a set of data.
- Designer bases like these can be much more efficient at representing functional variation.

Difference estimation of derivatives

- It is common practice to estimate a derivative by taking the difference between adjacent function values divided by the difference between adjacent time values:

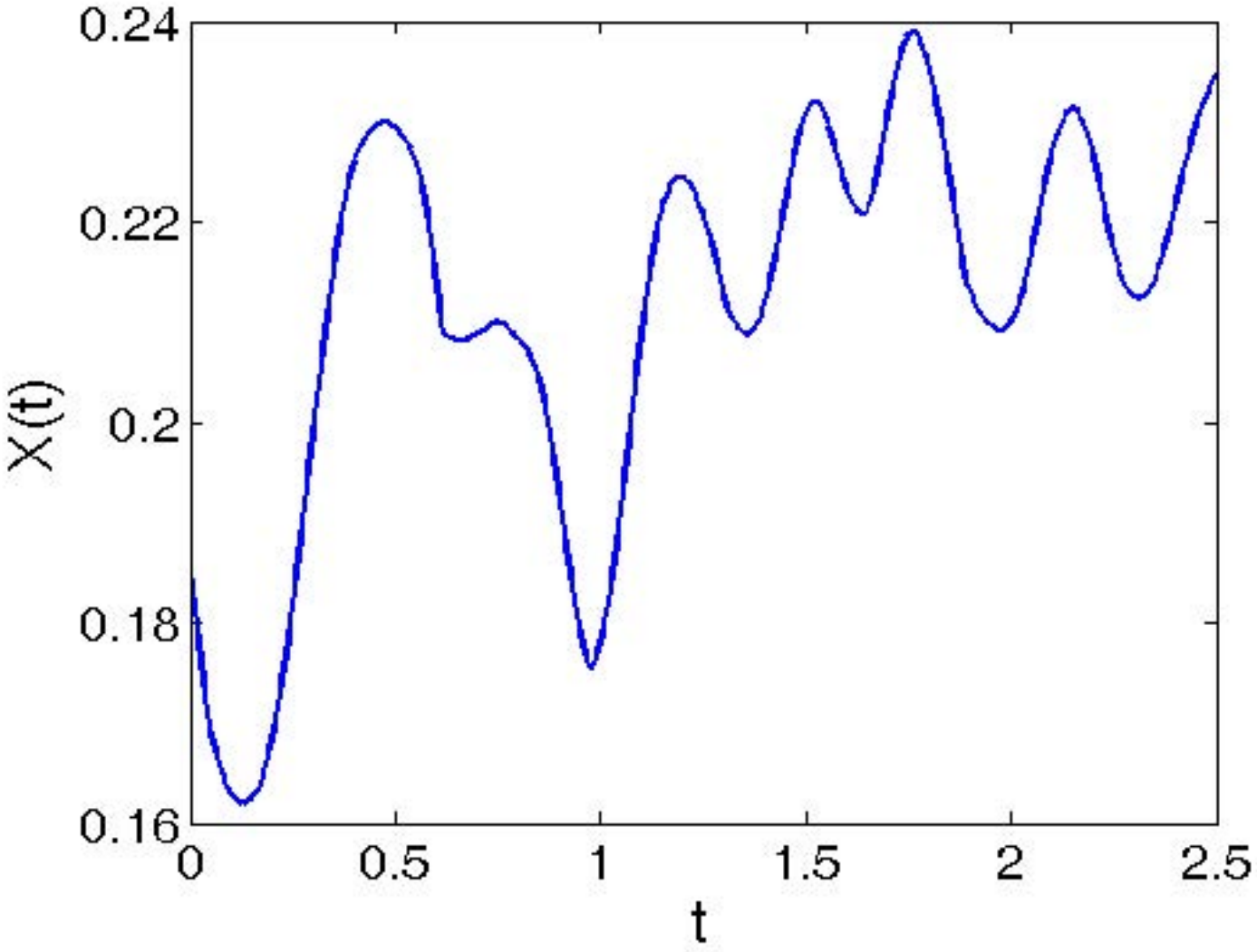
$$x'(t_i) = \frac{x(t_{i+1}) - x(t_i)}{t_{i+1} - t_i}.$$

- The second derivative can be estimated by applying this differencing process to the first difference ratios.
- This only works for very smooth functions observed without appreciable error.
- Even the smallest amount of error is greatly magnified by differencing.

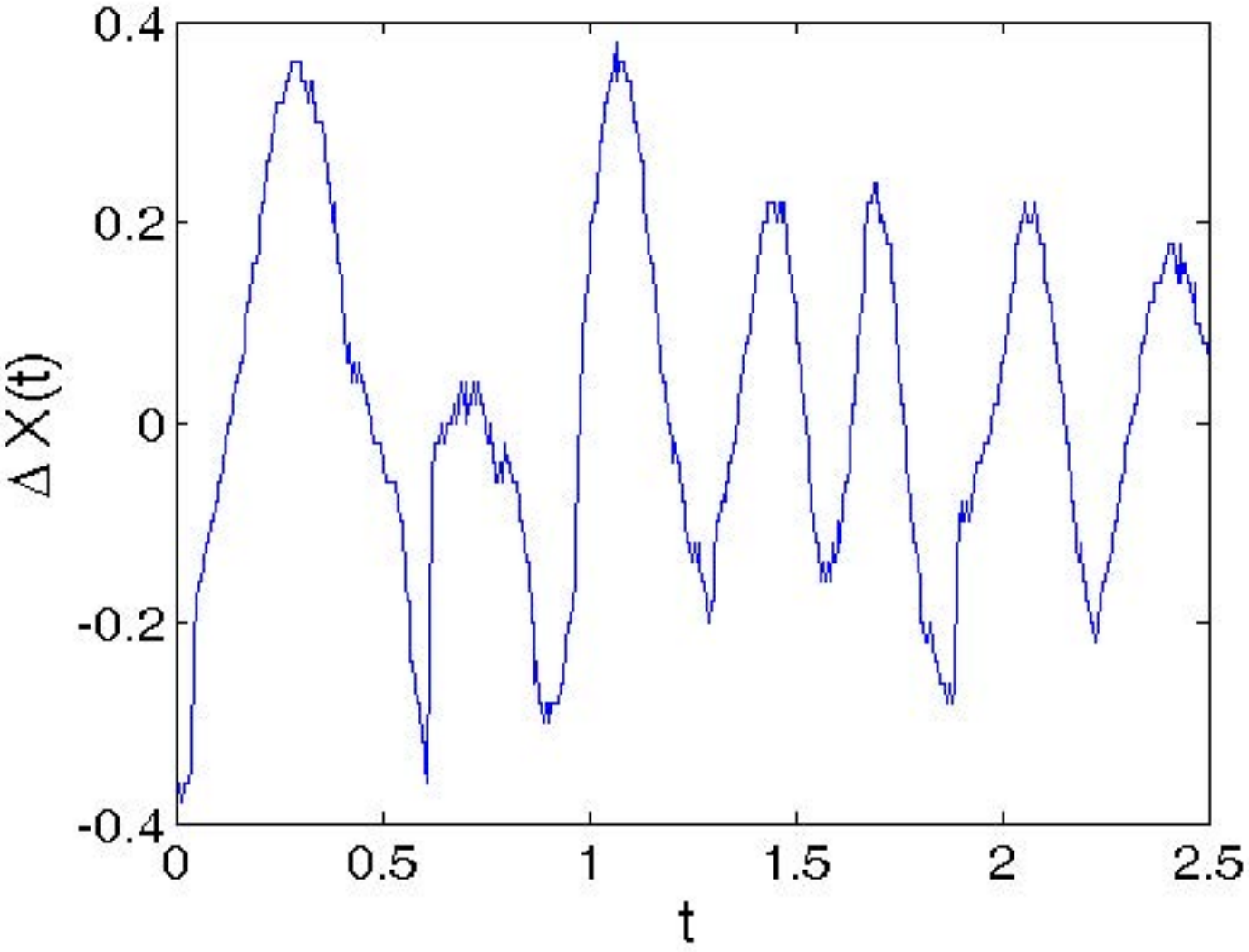
An example: pen position

- The following displays show what happens when we difference a record of pen position that has a signal-to-noise ratio of 500-to-1, and which is sampled 200 times per second.

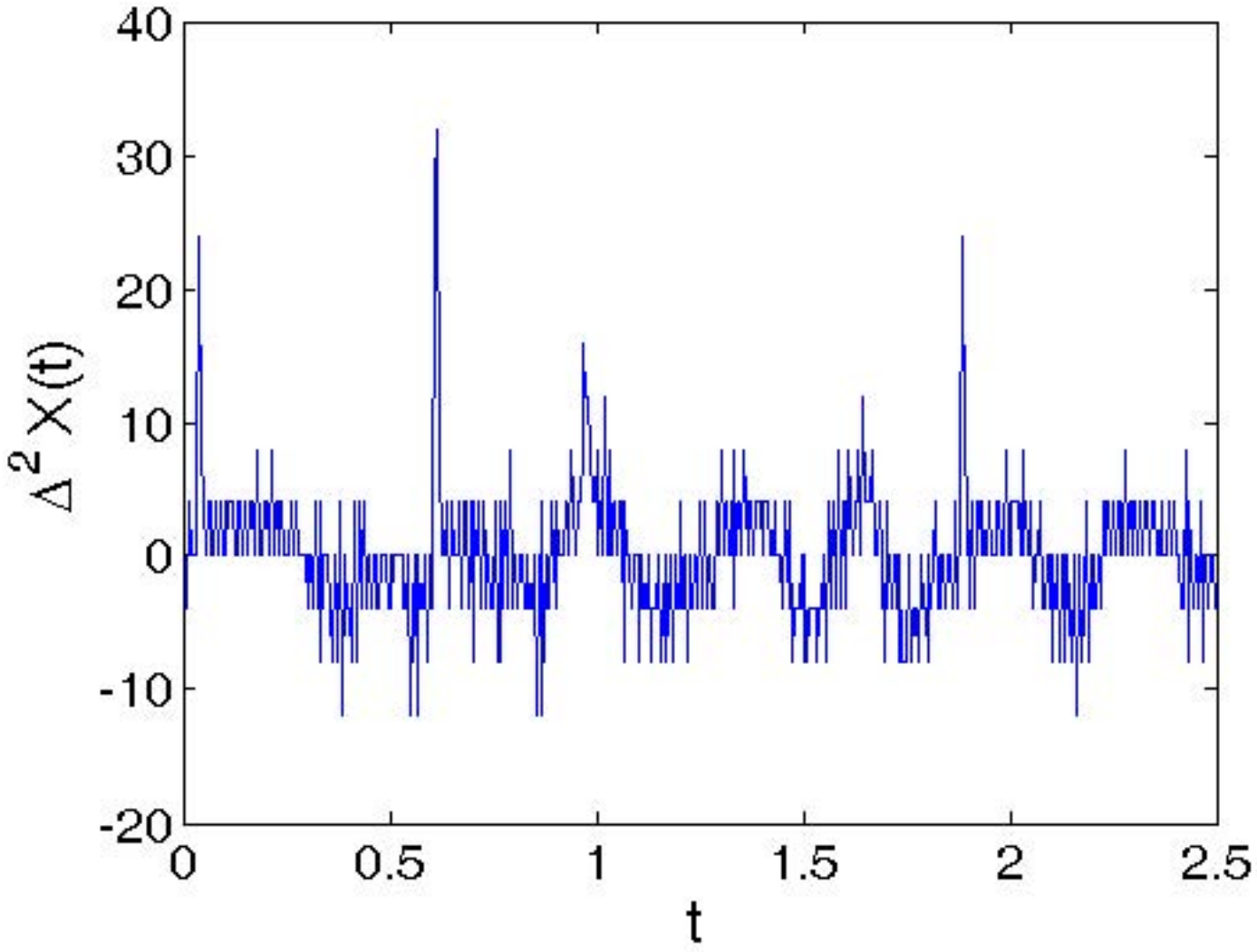
The pen position function



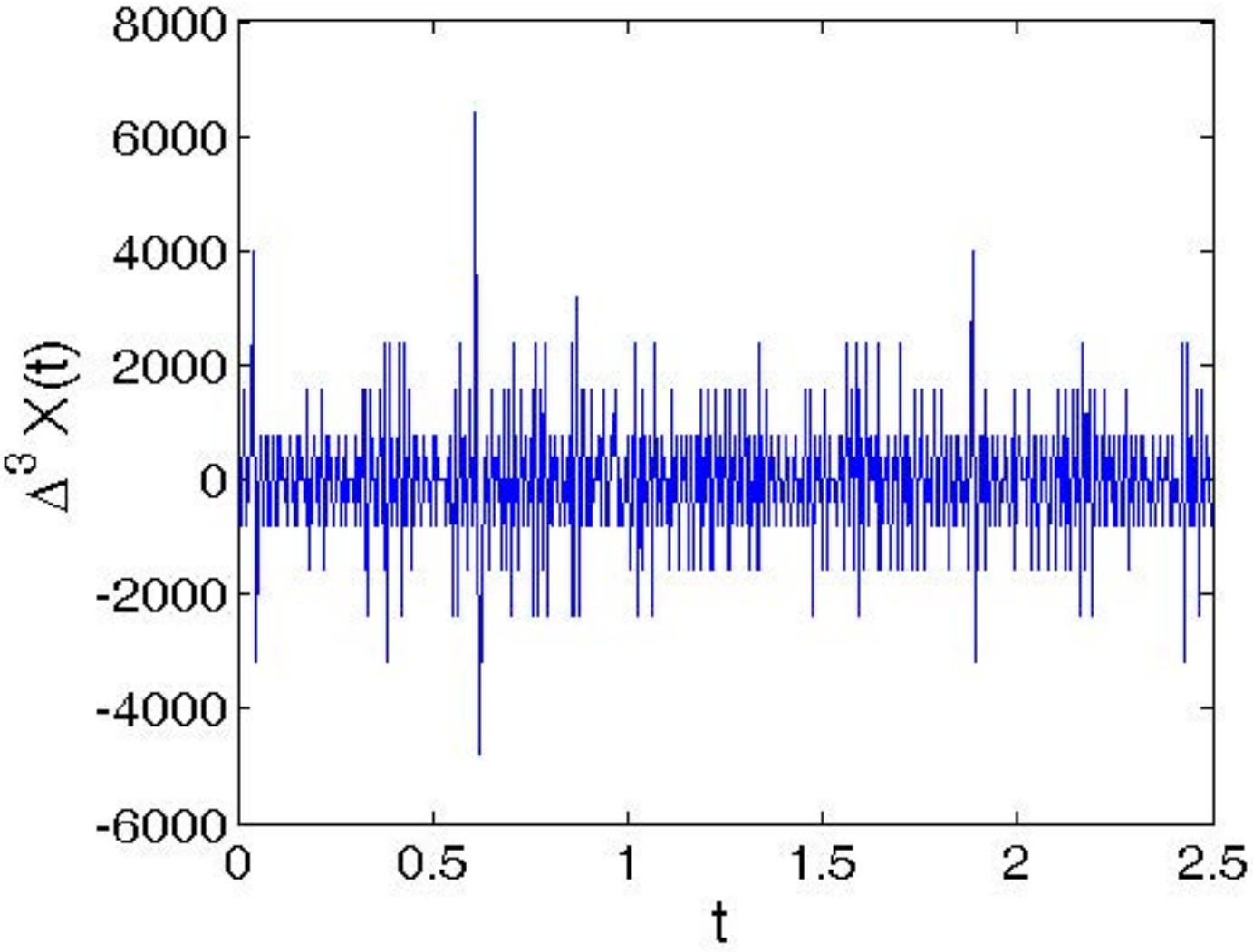
The first difference ratios



The second difference ratios



The third difference ratios



An example: pen position

- The third order difference ratios are virtually worthless as an estimates of the values of the third derivative, which we will need in later analyses.

Where do we go from here?

- Now we need to see how to fit a basis function expansion to noisy data.
- The simplest process is through least squares approximation (including kernel regression and local polynomials).
- This is essentially the use of multiple regression analysis, where the covariates are the basis function values corresponding to time sampling points.
- This works reasonably well, but we will see how to do even better later.